

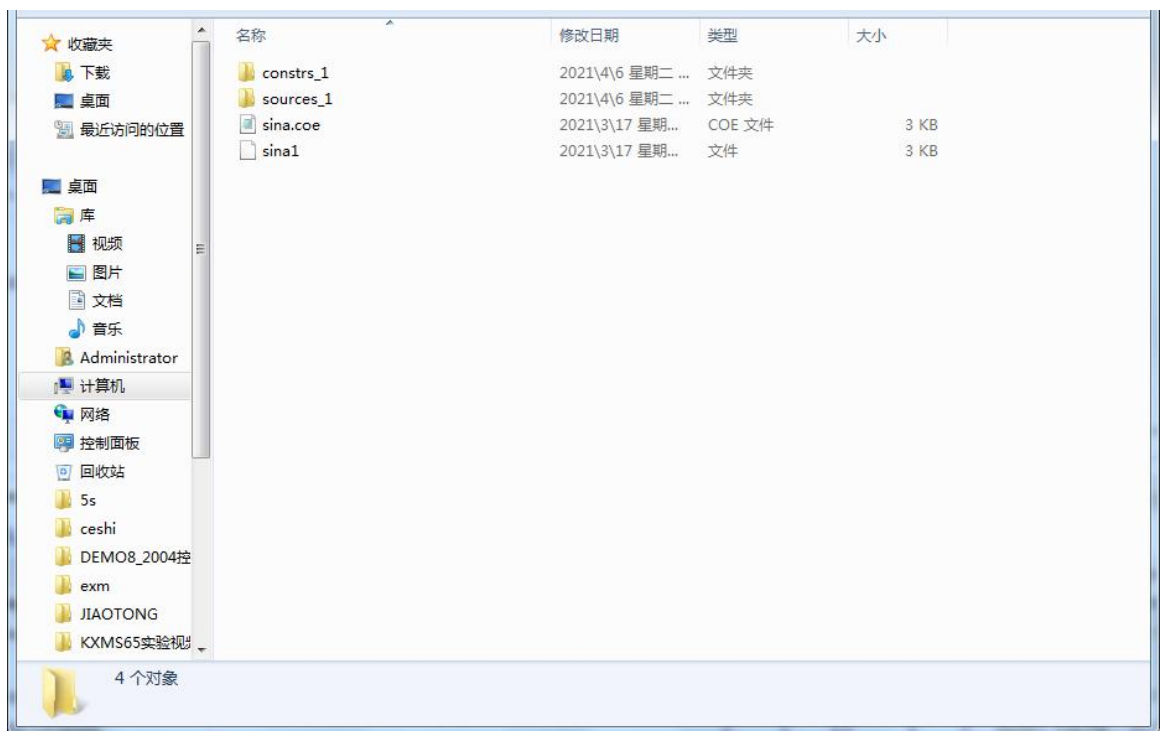
DA902 显示正弦波

1、实验目的：

制定 ROM 数据通过 12bitDA 产生正弦信号发生器。

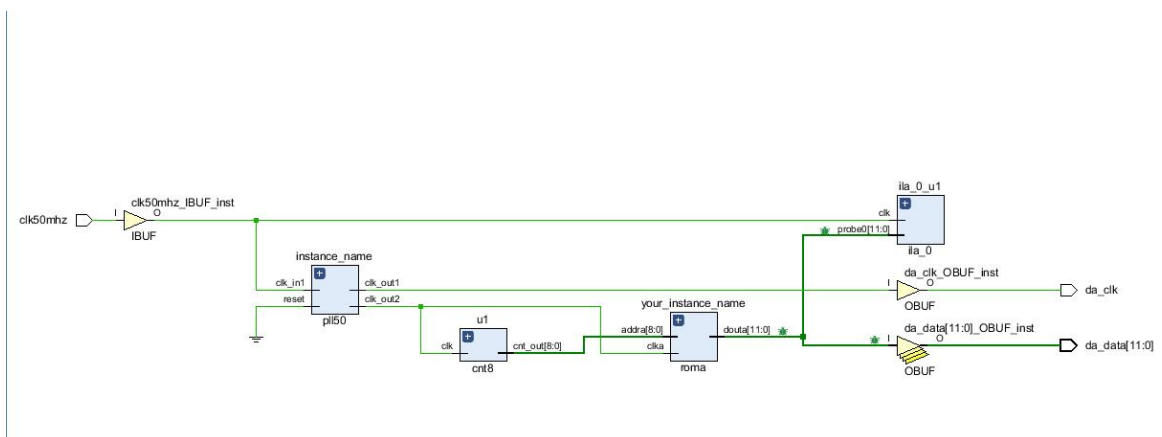
2、实验原理：

在 FPGA 内部定制一个正弦信号发生的数据，详见此例 SIN.COE 图 24.1，在送给 DA 转换成模拟信号，通过示波器观察。



24.1 生成的 coe 正弦波文件。

1. RTL 图文件如下 24.2

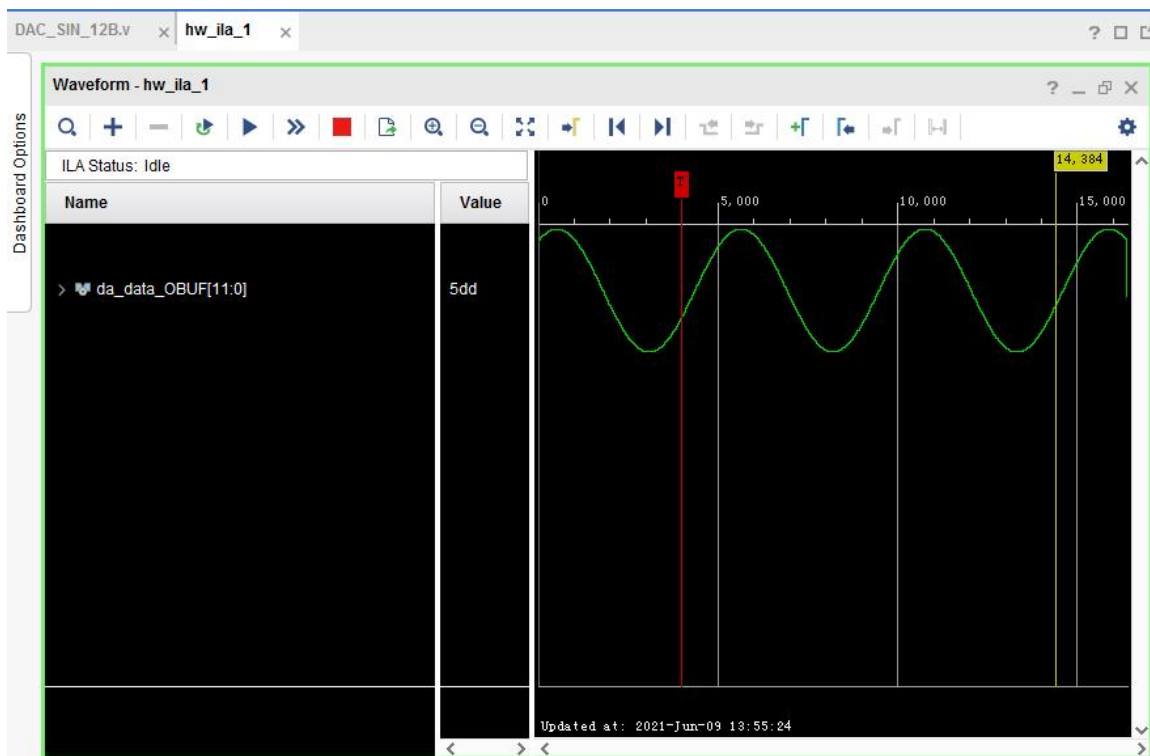


24.2 RTL 原理图

板载 50M 时钟经过锁相环，分别输出两路信号，一路倍频 100M，作为 DA 的输入时钟，一路分频输出 1MHz 和读取 ROM 地址控制器信号。

ROM 定制输入端是地址 9 位输入，内部嵌入定制好的 sina.coe 文件，根据每个地址的输入，读取 sina 里对应的数据，通过 12 位输出口输出，输出口接外围的 DA902，转换成模拟信号输出。接示波器观察波形。

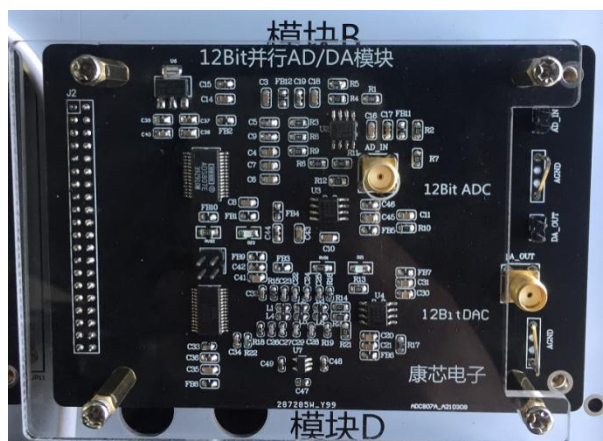
2. da 逻辑分析仪波形图 24-3:



24-3 逻辑分析仪实时波形

3、硬件准备:

图 24.4 上端是并行采样速率 50MHz 12 位 AD807 处理芯片，红色皮套是专用模拟信号采用端口，标有 AD_IN 是针式采用端口，AGND 是接地，图下端是并行转换速率 165MHz 12 位 DA902 处理芯片，红色皮套是专用模拟信号输出端口，标有 DA_IN 是针式输出端口，AGND 是接地。图 24.4 是 AD807 芯片的引脚图，图 24.5 是 DA902 芯片的引脚图，图 24.6，24.7 分别是 AD807 和 DA902 芯片接口原理图 AD_D0~AD_D11, AD_CLK 和 DA_D0~DA_D11 DA_CLK 分别是 AD807 和 DA902 数据脚和时钟输入脚。



24-4

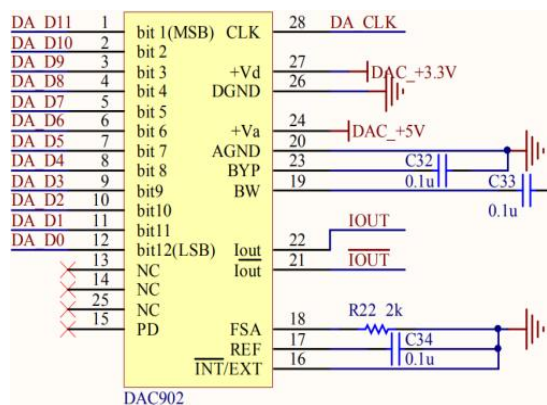


图 24-5 AD807 芯片原理图

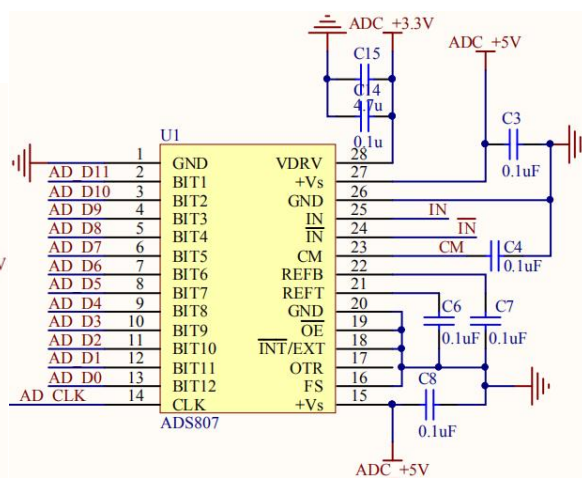


图 24-6 DA902 芯片原理图

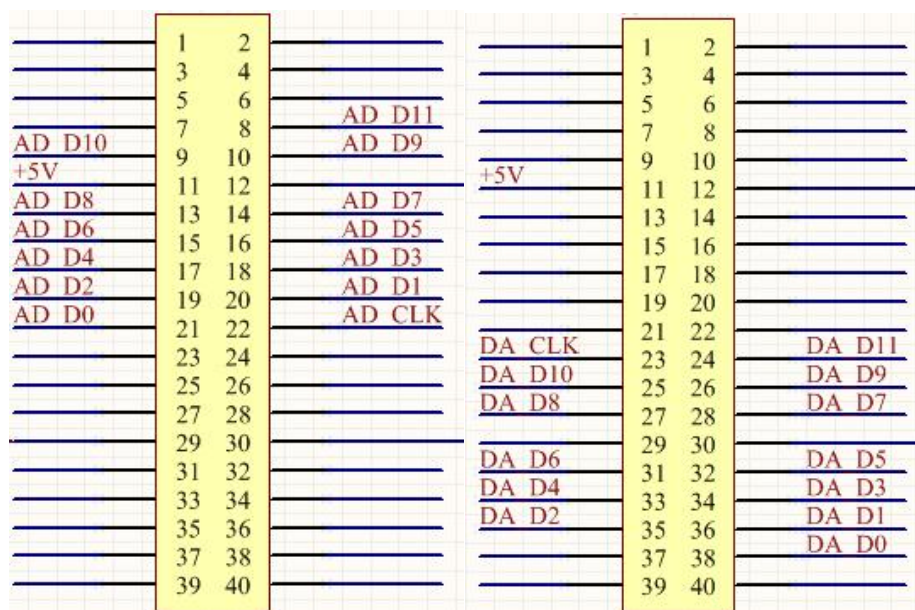


图 24-7 AD80740 芯接口引脚原理图

图 24-8 DA90240 芯接口引脚原理图

4、管脚约束：

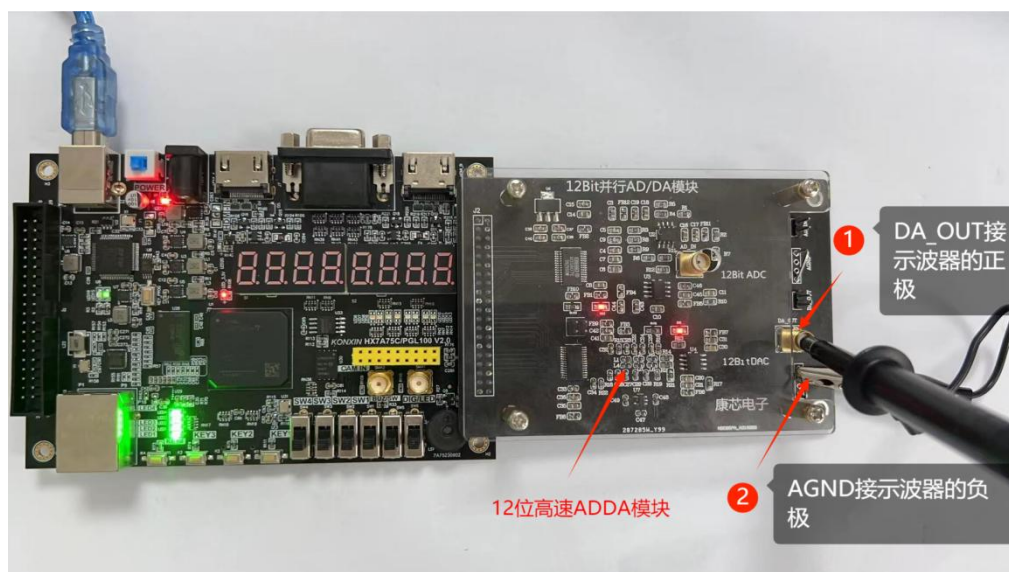
#IO 管脚约束

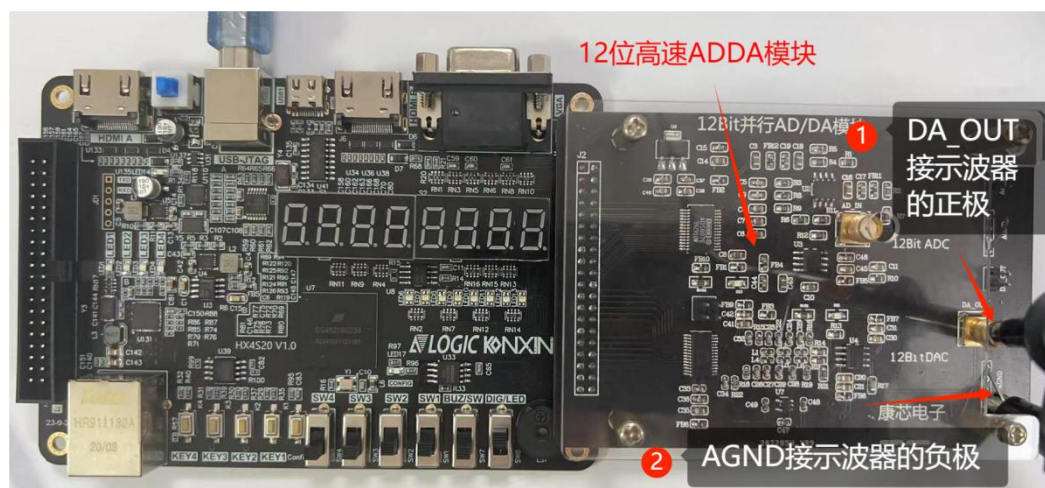
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set_pin_assignment { clk50mhz } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { da_clk } { LOCATION = K15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[0] } { LOCATION = R15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[10] } { LOCATION = H15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[11] } { LOCATION = J16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[1] } { LOCATION = P15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[2] } { LOCATION = P16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[3] } { LOCATION = N14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[4] } { LOCATION = N16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[5] } { LOCATION = L16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[6] } { LOCATION = M15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[7] } { LOCATION = M16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[8] } { LOCATION = K16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { da_data[9] } { LOCATION = H16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8; PULLTYPE = NONE; }

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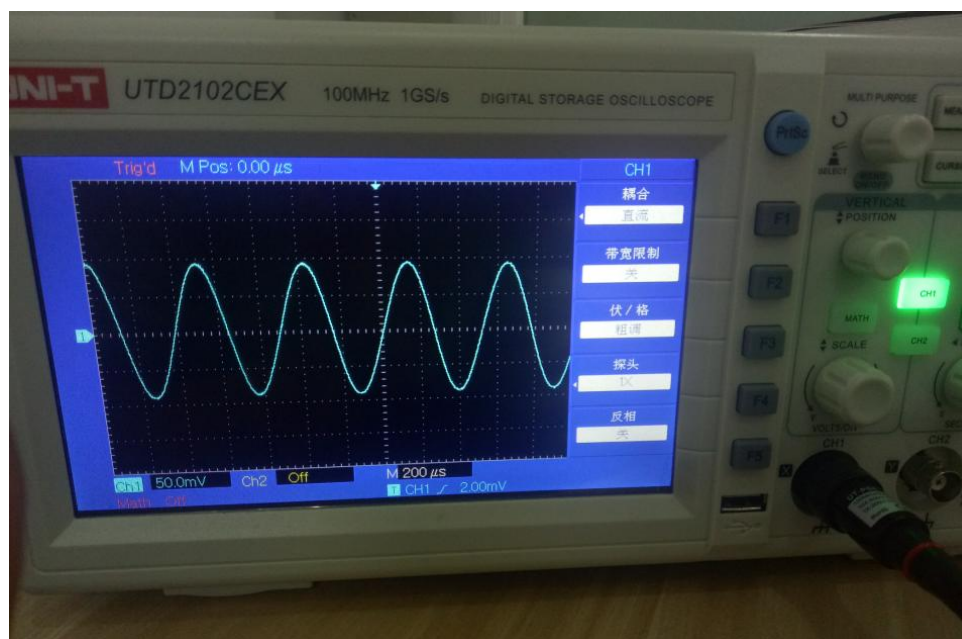
5、硬件连接





6、实验结果：

内部 ROM 描点产生的正弦信号



24-7 示波器显示 DA902 芯片显示的内容